

REFERENCES

A. Foundational TIF Biology

- [1] Wang L, Jin G, Zhou Q, Liu Y, Zhao X, Li Z, Yin N, Peng M. Induction of immortal-like and functional CAR T cells by defined factors. *J Exp Med*. 2024;221(5):e20232368. DOI: 10.1084/jem.20232368. PMC10965394.
- [2] Xu J, Jia Z, Zhao X, Wang L, Jin G, Li Z, Yin N, Li Y, Peng M. BCOR and ZC3H12A suppress a core stemness program in exhausted CD8+ T cells. *J Exp Med*. 2025;222(8):e20241133. DOI: 10.1084/jem.20241133. PMID: 40327039.
-

B. NIR-Driven c-di-GMP Optogenetic Module

- [3a] Ryu MH, Gomelsky M. Near-infrared light responsive synthetic c-di-GMP module for optogenetic applications. *ACS Synth Biol*. 2014;3(11):802–810. DOI: 10.1021/sb400182x. PMC4277780.
- [3b] Folcher M, Oesterle S, Zwicky K, Kastelic N, Daoud-El Baba M, Wagner JA, Geering B, Fussenegger M. Mind-controlled transgene expression by a wireless-powered optogenetic designer cell implant. *Nat Commun*. 2014;5:5392. DOI: 10.1038/ncomms6392. PMC4241983.
- [SAVI] David C, Frémond ML. Lung inflammation in STING-associated vasculopathy with onset in infancy (SAVI). *Cells*. 2022;11(3):318. DOI: 10.3390/cells11030318. PMID: 35159128. PMC8834229.
-

C. MrkH Bacterial Function and Operator Characterization

- [4a] Wilksch JJ, Yang J, Clements A, Dunstan RA, Thapa R, Little MA, Lai VC, May KL, Lithgow T, Strugnell RA. MrkH, a novel c-di-GMP-dependent transcriptional activator, controls *Klebsiella pneumoniae* biofilm formation by regulating type 3 fimbriae expression. *PLoS Pathog*. 2011;7(8):e1002204. DOI: 10.1371/journal.ppat.1002204. PMC3161979.
- [4b] Yang J, Wilksch JJ, Tan JWH, Dunstan RA, Webb CT, Lithgow T, Strugnell

RA, Robins-Browne RM. Identification of the molecular determinants of MrkH-driven gene regulation in *K. pneumoniae*. *J Bacteriol*. 2015;197(9):1659–1667. PMC4403657.

[4c] Ares MA, Fernández-Vázquez JL, Pacheco S, Martínez-Santos VI, Jarillo-Quijada MD, Torres J, Alcántar-Curiel MD, Girón JA, De la Cruz MA. Additional regulatory activities of MrkH for the transcriptional expression of the *Klebsiella pneumoniae* mrk genes: antagonist of H-NS and repressor. *PLoS One*. 2017;12(3):e0173285. DOI: 10.1371/journal.pone.0173285. PMC5344390.

D. Bone Piezoelectricity and Mechanotransduction

[5] Fukada E, Yasuda I. On the piezoelectric effect of bone. *J Phys Soc Japan*. 1957;12:1158–1162. DOI: 10.1143/JPSJ.12.1158.

[6] Li X, Han L, Nookaew I, Mannen E, Silva MJ, Almeida M, Xiong J. Stimulation of Piezo1 by mechanical signals promotes bone anabolism. *eLife*. 2019;8:e49631. DOI: 10.7554/eLife.49631. PMC6779475.

E. Vector Delivery and Integration

[7] Addgene Lentiviral Vector Guide. Transfer plasmid packaging capacity 8–10 kb. <https://www.addgene.org/guides/lentivirus/>. Accessed April 2026.

[8] Yamaguchi T, Seneviratne R, Seneviratne A, Chang DF. Efficient viral delivery of Cas9 into human safe harbor. *Sci Rep*. 2020;10(1):21391. DOI: 10.1038/s41598-020-78450-8. PMC7722726.

F. HO-1 Modulation

[9] Alaluf E, Vodo P, Schulz C. Heme oxygenase-1 orchestrates the immunosuppressive program of tumor-associated macrophages. *J Clin Invest*. 2020;130(10):5400–5416. DOI: 10.1172/jci.insight.133929. PMC7308058.

[10] Kim SJ, Kim MJ, Lee SH. Overexpression of HO-1 contributes to sepsis-induced immunosuppression by modulating the Th1/Th2 balance and

regulatory T-cell function. *J Infect Dis.* 2017;215(10):1608–1618. DOI: 10.1093/infdis/jix171. PMID: 28368519.

A-Series. Bleeding-Edge Biosensing

[A1] Yang Y, Gao W. Wearable and flexible electronics for continuous molecular monitoring. *Chem Soc Rev.* 2019;48(6):1465–1491. DOI: 10.1039/C7CS00730B.

[A2] Wang M, Yang Y, Min J, Song Y, Yu J, Mukasa D, Gao W. A wearable electrochemical biosensor for the monitoring of metabolites and nutrients. *Nat Biomed Eng.* 2022;6(11):1225–1235. DOI: 10.1038/s41551-022-00916-z.

[A3] Chen Y, He Z, Wu Y, Bai X, Li Y, Yang W, Liu Y, Li R-W. A wearable molecularly imprinted electrochemical sensor for cortisol stable monitoring in sweat. *Biosensors.* 2025;15(3):194. DOI: 10.3390/bios15030194.

[A4] Garg SK, Kipnes M, Castorino K, Bailey TS, Conget I, Welsh JB, PROMISE Study Group. Evaluation of accuracy and safety of the next-generation up to 180-day long-term implantable Eversense continuous glucose monitoring system: The PROMISE Study. *Diabetes Technol Ther.* 2022;24(2):84–92. DOI: 10.1089/dia.2021.0182.

[A5] Suryaprabha T, Choi C, Hwang B. Smart wearable and implantable biosensors for continuous health monitoring: materials, biocompatibility, and AI integration. *npj Flex Electron.* 2026;10:46. DOI: 10.1038/s41528-026-00560-6.

B-Series. Osteoimmunology and Immune-Mediated Osseointegration

[B1] Yun JI, Yun SI, Kim JH, Kim DG, Lee DW. Mediation of osseointegration, osteoimmunology, and osteoimmunologic integration by Tregs and macrophages: a narrative review. *Int J Mol Sci.* 2025;26(11):5421. PMC12154945.

[B2] Vandamme K, Holy X, Bensidhoum M, Logeart-Avramoglou D, Naert IE,

Duyck JA, Petite H. In vivo molecular evidence of delayed titanium implant osseointegration in compromised bone. *Biomaterials*. 2011;32(14):3547–3554. DOI: 10.1016/j.biomaterials.2011.01.056.

[B3] Kim JS, Kang SM, Seo KW, Nahm KY, Chung KR, Kim SH, Ahn JP. Nanoscale bonding between human bone and titanium surfaces: osseohybridization. *Biomed Res Int*. 2015;2015:960410. DOI: 10.1155/2015/960410. PMC4312565.

[B4] Kumar PS, KS SK, Grandhi VV, Gupta V. The effects of titanium implant surface topography on osseointegration: literature review. *JMIR Biomed Eng*. 2019;4(1):e13237. DOI: 10.2196/13237.

C-Series. Biomineralization and Biosynthetic Material Deployment

[C1] Faivre D, Schüler D. Magnetosome biomineralization by magnetotactic bacteria. In: Schüler D, ed. *Microbiology Monographs*, Vol. 23. Springer, Cham; 2022. DOI: 10.1007/978-3-030-80807-5_7.

[C2] Lin W, Pan Y, Bazylinski DA et al. Key gene networks that control magnetosome biomineralization in magnetotactic bacteria. *Natl Sci Rev*. 2023;10(1):nwac238. DOI: 10.1093/nsr/nwac238.

[C3] Siponen MI, Legrand P, Widdrat M, Jones SR, Zhang W-J, Chang MCY, Faivre D, Schüler D. Structural insight into magnetochrome-mediated magnetite biomineralization. *Nature*. 2013;502(7473):681–684. DOI: 10.1038/nature12573.

[C4] Amor M, Komeili A. Magnetochrome-catalyzed oxidation of ferrous iron by MamP enables magnetite crystal growth in the magnetotactic bacterium AMB-1. *Proc Natl Acad Sci USA*. 2024;121(14):e2410245121. DOI: 10.1073/pnas.2410245121.

[C5] Yamagishi A, Tanaka M, Matsunaga T. Control of magnetite nanocrystal morphology in magnetotactic bacteria by regulation of mms7 gene expression. *Sci Rep*. 2016;6:29785. DOI: 10.1038/srep29785.

[C6] Wan J, Ji R, Liu J, Ma K, Pan Y, Lin W. Biomineralization in magnetotactic bacteria: from diversity to molecular discovery-based applications. *Cell Rep*.

2024;43(12):114995. DOI: 10.1016/j.celrep.2024.114995.

[C7] Zhou Y, Aslan H, Dong M et al. Engineered bacteria that self-assemble bioglass polysilicate coatings display enhanced light focusing. *bioRxiv*. 2024. DOI: 10.1101/2024.06.03.597164.

[C8] Vigil TN, Schwendeman NK, Grogger MLM, Morrison VL, Warner MC, Bone NB, Vance MT, Morris DC, McElmurry K, Berger BW, Steel JJ. Surface-displayed silicatein- α enzyme in bioengineered *E. coli* enables biocementation and silica mineralization. *Front Syst Biol*. 2024. DOI: 10.3389/fsysb.2024.1377188.

[C9] Zhou Y, Li W, Qi J et al. Directed evolution of silicatein reveals biomineralization synergism between protein sequences. *ACS Omega*. 2024. DOI: 10.1021/acsomega.4c06359.

[C10] Kotzsch A, Pawolski D, Milentyev A, Shevchenko A, Scheffel A, Poulsen N, Shevchenko A, Kröger N. Silicanin-1 is a conserved diatom membrane protein involved in silica biomineralization. *BMC Biol*. 2017;15:65. DOI: 10.1186/s12915-017-0400-8.

[C11] Li H, Gao H, Wu Q, Adam V, Kuca K. Recombinant ferritins for multimodal nanomedicine. *J Enzyme Inhib Med Chem*. 2023;38(1):2219868. DOI: 10.1080/14756366.2023.2219868. PMC10236959.

[C12] Cao S, Dhungana M, Shengjuler J, Zhu M. Self-assembled ferritin nanoparticles for delivery of antigens and development of vaccines: from structure and property to applications. *Molecules*. 2024;29(17):4221. DOI: 10.3390/molecules29174221. PMC11397193.

[C13] Radkevich A, Srivastava A, Arroyo P, Matalan A. Ferritin nanocages as efficient nanocarriers and promising platforms for COVID-19 and other vaccines development. *Biochim Biophys Acta Gen Subj*. 2023;1867(3):130288. DOI: 10.1016/j.bbagen.2022.130288. PMC9721431.

[C14] Liu X, Lopez PA, Giessen TW, Giles M, Way JC, Silver PA. Engineering genetically-encoded mineralization and magnetism via directed evolution. *Sci Rep*. 2016;6:38019. DOI: 10.1038/srep38019. PMC5126674.

[C15] Sigmund F, Massner C, Erdmann P, Stelzl A, Rolbieski H, Desai M, Bricault S, Wörner TP, Snijder J, Geerts A, Fuchs H, Hrabě de Angelis M, Beck J, Ntziachristos V, Plitzko J, Westmeyer GG. Bacterial encapsulins as

orthogonal compartments for mammalian cell engineering. *Nat Commun.* 2018;9:1990. DOI: 10.1038/s41467-018-04227-3. PMC5959871.

[C16] Maity B, Abe S, Ueno T. Observation of gold sub-nanocluster nucleation within a protein cage. *Nat Commun.* 2017;8:14820. DOI: 10.1038/ncomms14820. PMC5357307.

[C17] Zhou Z, Luo Q, Zheng L, Xu J. Design of a gold clustering site in an engineered apo-ferritin cage. *Commun Chem.* 2022;5:39. DOI: 10.1038/s42004-022-00651-1. PMC9814837.

D-Series. Wireless Power Transfer for Medical Implants

[D1] Agarwal K, Jegadeesan R, Guo Y-X, Thakor NV. Wireless power transfer strategies for implantable bioelectronics. *IEEE Rev Biomed Eng.* 2017;10:136–161. DOI: 10.1109/RBME.2017.2683520. PMC7349694.

[D2] Khan SR, Pavuluri SK, Cummins G, Desmulliez MPY. Wireless power transfer techniques for implantable medical devices: a review. *Sensors.* 2020;20(12):3487. DOI: 10.3390/s20123487.

[D3] Kim J, Park S, Lee S, Cho IK, Kim J, Lee H. Wireless power transfer modalities for implantable bioelectronics: electromagnetic, acoustic, and magneto-dynamic perspectives. *Adv Mater Technol.* 2026. DOI: 10.1002/admt.202502002.

[D4] Soltani N, Aliroteh MS, Salam MT, Velazquez JLP, Genov R. Extending the limits of wireless power transfer to miniaturized implantable electronic devices. *Bioengineering.* 2018;5(4):84. DOI: 10.3390/bioengineering5040084. PMC6187913.

[D5] IEEE Std C95.1-1999. IEEE Standard for Safety Levels with Respect to Human Exposure to Electric, Magnetic, and Electromagnetic Fields, 0 Hz to 300 GHz. Institute of Electrical and Electronics Engineers; 1999.

[D6] IEEE Std C95.1-2019. IEEE Standard for Safety Levels with Respect to Human Exposure to Electric, Magnetic, and Electromagnetic Fields, 0 Hz to 300 GHz. Institute of Electrical and Electronics Engineers; 2019. ISBN 978-1-5044-5548-0.

[D7] IEC 60601-2-33:2010+AMD1:2013+AMD2:2015. Medical Electrical Equipment — Part 2-33: Particular requirements for the basic safety and essential performance of magnetic resonance equipment for medical diagnosis. International Electrotechnical Commission; Geneva.

[D8] ICNIRP. Guidelines for limiting exposure to time-varying electric, magnetic and electromagnetic fields (up to 300 GHz). *Health Phys.* 1998;74(4):494–522.

[D9] Carmichael DW, Thornton JS, Rodionov R, Thornton R, McEvoy AW, Allen PJ, Lemieux L, Rosenkranz M. Safety of localizing epilepsy monitoring intracranial electroencephalograph electrodes using MRI: radiofrequency-induced heating. *J Magn Reson Imaging.* 2008;28(5):1233–1244. DOI: 10.1002/jmri.21583. PMC2883075.

[D10] Federal Communications Commission. Evaluating Compliance with FCC Guidelines for Human Exposure to Radiofrequency Electromagnetic Fields. OET Bulletin 65, Edition 97-01. FCC Office of Engineering and Technology; 1997.

E-Series. Mitochondrial Bioenergetics

[E1] Zhao R-Z, Jiang S, Zhang L, Yu Z-B. Mitochondrial electron transport chain, ROS generation and uncoupling (Review). *Int J Mol Med.* 2019;44(1):3–15. DOI: 10.3892/ijmm.2019.4188. PMC6559295.

[E2] Vasan K, Clutter M, Fernandez Dunne S, George MD, Luan C-H, Chandel NS, Martínez-Reyes I et al. Genes involved in maintaining mitochondrial membrane potential upon electron transport chain disruption. *Front Cell Dev Biol.* 2022;10:781558. DOI: 10.3389/fcell.2022.781558. PMC8888678.

[E3] Vogt S, Rhiel A, Weber P, Ramzan R. Revisiting Kadenbach: electron flux rate through cytochrome c-oxidase determines the ATP-inhibitory effect and subsequent production of ROS. *BioEssays.* 2017;39(4):e201600043. DOI: 10.1002/bies.201600043. PMC5084804.

F-Series. Microbial Extracellular Electron Transfer

- [F1] Hu Y, Wang Y, Han X, Shan Y, Li F, Shi L et al. Biofilm biology and engineering of *Geobacter* and *Shewanella* spp. for energy applications. *Front Bioeng Biotechnol.* 2021;9:786416. DOI: 10.3389/fbioe.2021.786416. PMC8683041.
- [F2] Walker DJF, Adhikari RY, Holmes DE, Ward JE, Woodard TL, Nevin KP, Lovley DR. Electrically conductive pili from pilin genes of phylogenetically diverse microorganisms. *ISME J.* 2018;12(1):48-58. DOI: 10.1038/ismej.2017.141. PMC5739001.
- [F3] Holmes DE, Dang Y, Walker DJF, Lovley DR. The electrically conductive pili of *Geobacter* species are a recently evolved feature for extracellular electron transfer. *Microb Genomics.* 2016;2(8):e000072. DOI: 10.1099/mgen.0.000072. PMC5320591.
- [F4] Gralnick JA, Bond DR. Electron transfer beyond the outer membrane: putting electrons to rest. *Annu Rev Microbiol.* 2023;77:517-539. DOI: 10.1146/annurev-micro-032221-023725.
-

G-Series. PGC-1 α and Mitochondrial Biogenesis

- [G1] Abu Shelbayeh O, Arroum T, Morris S, Busch KB. PGC-1 α is a master regulator of mitochondrial lifecycle and ROS stress response. *Antioxidants.* 2023;12(6):1075. DOI: 10.3390/antiox12051075. PMC10215733.
- [G2] Lin J, Wu H, Tarr PT, Zhang C-Y, Wu Z, Boss O, Michael LF, Puigserver P, Isotani E, Olson EN, Lowell BB, Bassel-Duby R, Spiegelman BM. Transcriptional co-activator PGC-1 α drives the formation of slow-twitch muscle fibres. *Nature.* 2002;418(6899):797-801. DOI: 10.1038/nature00904.
- [G3] Lehman JJ, Barger PM, Kovacs A, Saffitz JE, Medeiros DM, Kelly DP. Peroxisome proliferator-activated receptor gamma coactivator-1 promotes cardiac mitochondrial biogenesis. *J Clin Invest.* 2000;106(7):847-856. DOI: 10.1172/JCI10268.
- [G4] Fernandez-Marcos PJ, Auwerx J. Regulation of PGC-1 α , a nodal regulator of mitochondrial biogenesis. *Am J Clin Nutr.* 2011;93(4):884S-890S. DOI: 10.3945/ajcn.110.001917.

H-Series. AMPK Energy Sensing

[H1] Hardie DG, Ross FA, Hawley SA. AMPK: a nutrient and energy sensor that maintains energy homeostasis. *Nat Rev Mol Cell Biol.* 2012;13(4):251–262. DOI: 10.1038/nrm3311.

[H2] Jäger S, Handschin C, St-Pierre J, Spiegelman BM. AMP-activated protein kinase (AMPK) action in skeletal muscle via direct phosphorylation of PGC-1alpha. *Proc Natl Acad Sci USA.* 2007;104(29):12017–12022. DOI: 10.1073/pnas.0705070104.

[H3] Zong H, Ren JM, Young LH, Pypaert M, Mu J, Birnbaum MJ, Shulman GI. AMP kinase is required for mitochondrial biogenesis in skeletal muscle in response to chronic energy deprivation. *Proc Natl Acad Sci USA.* 2002;99(25):15983–15987. DOI: 10.1073/pnas.252625599.

[H4] Kola B. AMPK as a mediator of hormonal signalling. *J Endocrinol.* 2008;199(1):1–11. DOI: 10.1677/JME-09-0063.

M-Series. Bioheat Modeling and Thermal Safety

[M3] Pennes HH. Analysis of tissue and arterial blood temperatures in the resting human forearm. *J Appl Physiol.* 1948;1(2):93–122. DOI: 10.1152/jappl.1948.1.2.93.

[M4] Neufeld E, Gosselin B, Ghaderi H, Kuster N et al. Modeling for neurosurgical laser interstitial thermal therapy with and without intracranial recording electrodes. *Curr Res Neurosci.* 2024. DOI: 10.1016/j.crneur.2024.100139.

[M5] Zhao Y, Tang L, Rennaker R, Hutchens C, Soo Lee B, McIntyre CC. Studies in RF power communication, SAR, and temperature elevation in wireless implantable neural interfaces. *PLoS ONE.* 2013;8(11):e77759. DOI: 10.1371/journal.pone.0077759. PMC3819346.

[M6] Carmichael DW, Thornton JS, Rodionov R, Thornton R, McEvoy AW, Allen PJ, Lemieux L, Rosenkranz M. Safety of localizing epilepsy monitoring intracranial electroencephalograph electrodes using MRI: radiofrequency-

induced heating. *J Magn Reson Imaging*. 2008;28(5):1233–1244. DOI: 10.1002/jmri.21583. PMC2883075.

[M7] Memezawa H, Minamisawa H, Smith M-L, Siesjö BK. Delayed postischemic hyperthermia in awake rats worsens the histopathological outcome of transient focal cerebral ischemia. *Stroke*. 1996;27(12):2274–2281. DOI: 10.1161/01.STR.27.12.2274.

N-Series. Watchdog Timer and Fail-Safe Engineering

[N1] Koopman P. Proper watchdog timer use. *Better Embedded System SW [blog]*. Carnegie Mellon University; 2014. <https://betterembsw.blogspot.com/2014/05/proper-watchdog-timer-use.html>

[N2] Arshon Inc. Medical device design: principles, regulations, and safe electronics integration [technical article]; 2025. <https://arshon.com/blog/medical-device-design-principles-regulations-and-safe-electronics-integration/>

[N3] Bafna A, Verma S. FPGA based fault recovery architecture of watchdog timer. *Int J Res Publ Rev*. 2025;6(9):1595–1604. <https://ijrpr.com/uploads/V6ISSUE9/IJRPR2025090901.pdf>

O-Series. Hypermetabolism Pathology

[O1] Williams FN, Herndon DN, Jeschke MG. The hypermetabolic response to burn injury and interventions to modify this response. *Clin Plast Surg*. 2009;36(4):583–596. DOI: 10.1016/j.cps.2009.05.001. PMC3776603.

[O2] Klein I, Ojamaa K. Thyroid hormone and the cardiovascular system. *N Engl J Med*. 2001;344(7):501–509. DOI: 10.1056/NEJM200102153440707.

[O3] Ertek S, Cicero AF. Hyperthyroidism and cardiovascular complications: a narrative review on the basis of pathophysiology. *Arch Med Sci*. 2013;9(5):944–952. DOI: 10.5114/aoms.2013.38685. PMC3832836.

[O4] Jeschke MG, Chinkes DL, Finnerty CC, Kulp G, Suman OE, Norbury WB, Branski LK, Gauglitz GG, Mlcak RP, Herndon DN. Burn size determines the inflammatory and hypermetabolic response. *Crit Care*. 2007;11(4):R90. DOI: 10.1186/cc6102. PMID: 17716366.

[O5] Williams FN, Herndon DN, Jeschke MG. The hypermetabolic response to burn injury and interventions to modify this response. *Clin Plast Surg.* 2009;36(4):583-596. DOI: 10.1016/j.cps.2009.05.001. PMC3776603.

P-Series. Implantable Medical Device Watchdog Architectures

[P1] Sieracki JM, North RB. Failsafe programming of implantable medical devices. US Patent 7,899,546. Assignee: Medtronic, Inc.; 2011.

[P2] Sieracki JM, North RB. Failsafe programming of implantable medical devices. US Patent 7,373,206. Assignee: Medtronic, Inc.; 2008.

[P3] Sieracki JM, North RB. Failsafe programming of implantable medical devices. US Patent 9,867,992. Assignee: Medtronic, Inc.; 2018.

[P4] Arshon Inc. Medical device design: principles, regulations, and safe electronics integration [technical article]; 2025. <https://arshon.com/blog/medical-device-design-principles-regulations-and-safe-electronics-integration/>

Q-Series. Synthetic Biology Gene Circuits

[Q1] Gardner TS, Cantor CR, Collins JJ. Construction of a genetic toggle switch in *Escherichia coli*. *Nature.* 2000;403(6767):339-342. DOI: 10.1038/35002131.

[Q2] Lohmueller J, Corr PB, Tucker CL. Progress toward the construction of a tri-stable genetic toggle switch in *E. coli*. *BMC Syst Biol.* 2007;1(Suppl 1):P39. DOI: 10.1186/1752-0509-1-S1-P39.

[Q3] Kassam T, Zhang W, Yu C, Zalewski C, Kiwimaegi K, Weiss R, Antunes MS, Prasad A, Medford J. A genetic toggle switch in plants. *ACS Synth Biol.* 2025. DOI: 10.1021/acssynbio.4c00777.

[Q4] Elowitz MB, Leibler S. A synthetic oscillatory network of transcriptional regulators. *Nature.* 2000;403(6767):335-338. DOI: 10.1038/35002125.

[Q5] Potvin-Trottier L, Lord ND, Vinnicombe G, Paulsson J. Synchronous long-term oscillations in a synthetic gene circuit. *Nature*. 2016;538(7626):514–517. DOI: 10.1038/nature19841.

[Q6] Tyler J, Shiu A, Walton J. Revisiting a synthetic intracellular regulatory network that exhibits oscillations. *arXiv*. 2018. <https://arxiv.org/abs/1808.00595>

R-Series. Treg Induction and Peripheral Tolerance via TGF- β /IL-10

[R1] Akdis M, Akdis CA. Regulatory T cells and immune regulation of allergic diseases: roles of IL-10 and TGF- β . *Genes Immun*. 2014;15(8):511–520. DOI: 10.1038/gene.2014.45.

[R2] Taylor A, Verhagen J, Blaser K, Akdis M, Akdis CA. Mechanisms of immune suppression by interleukin-10 and transforming growth factor- β : the role of T regulatory cells. *Immunology*. 2006;117(4):433–442. DOI: 10.1111/j.1365-2567.2006.02321.x. PMC1782242.

[R3] Li MO, Flavell RA. Contextual regulation of inflammation: a duet by transforming growth factor- β and interleukin-10. *Immunity*. 2008;28(4):468–476. DOI: 10.1016/j.immuni.2008.03.003.

[R4] Chen W, Jin W, Hardegen N, Lei K-J, Li L, Marinos N, McGrady G, Wahl SM. Conversion of peripheral CD4+CD25– naive T cells to CD4+CD25+ regulatory T cells by TGF- β induction of transcription factor Foxp3. *J Exp Med*. 2003;198(12):1875–1886. DOI: 10.1084/jem.20030152.

[R5] Roncarolo MG, Bacchetta R, Bordignon C, Narula S, Levings MK. Type 1 T regulatory cells. *Immunol Rev*. 2001;182:68–79. DOI: 10.1034/j.1600-065X.2001.1820107.x.

[R6] Hippen KL, Merkel SC, Schirm DK, Sieben CM, Sumstad D, Kadidlo DM, McKenna DH, Bromberg JS, Levine BL, Riley JL, June CH, Scheinberg P, Douek DC, Miller JS, Wagner JE, Blazar BR. Generation and large-scale expansion of human inducible regulatory T cells that suppress graft-versus-host disease. *Am J Transplant*. 2011;11(6):1148–1157. DOI: 10.1111/j.1600-6143.2011.03558.x. PMC4757416.

[R7] Proietto AI, van Dommelen S, Zhou P, Rizzitelli A, D’Amico A, Steptoe RJ,

Naik SH, Lahoud MH, Liu Y, Zheng P, Shortman K, Wu L. Dendritic cells in the thymus contribute to T-regulatory cell induction. *Proc Natl Acad Sci USA*. 2008;105(50):19869–19874. DOI: 10.1073/pnas.0810268105. PMC3125210.

W-Series. Atomically Precise Gold Nanocluster Electronics

[W1] Maity B, Abe S, Ueno T. Observation of gold sub-nanocluster nucleation within a protein cage. *Nat Commun*. 2017;8:14820. DOI: 10.1038/ncomms14820. PMC5357307.

[W2] Zhou Z, Luo Q, Zheng L, Xu J. Design of a gold clustering site in an engineered apo-ferritin cage. *Commun Chem*. 2022;5:39. DOI: 10.1038/s42004-022-00651-1. PMC9814837.

Y-Series. NV Diamond Quantum Computing Platform

[Y1] Balasubramanian G, Neumann P, Twitchen D, Markham M, Kolesov R, Mizuochi N, Isoya J, Achard J, Beck J, Tissler J, Jacques V, Hemmer PR, Jelezko F, Wrachtrup J. Ultralong spin coherence time in isotopically engineered diamond. *Nat Mater*. 2009;8(5):383–387. DOI: 10.1038/nmat2420.

[Y2] Shim JH, Niemeyer I, Zhang J, Suter D. Room-temperature high-speed nuclear-spin quantum memory in diamond. *Phys Rev A*. 2013;87(1):012301. DOI: 10.1103/PhysRevA.87.012301.

[Y3] Kim J, Tsugawa K, Ishihara M, Koga Y, Hasegawa M. Nanocrystalline diamond film growth on plastic substrates at temperatures below 100°C from low-temperature plasma. *Phys Rev B*. 2010;82:125460. DOI: 10.1103/PhysRevB.82.125460.

[Y4] Teraji T, Shinei C, Masuyama Y, Miyakawa M, Taniguchi T. Nitrogen concentration control during diamond growth for NV⁻ centre formation. *Philos Trans A Math Phys Eng Sci*. 2024;382(2265):20220322. DOI: 10.1098/rsta.2022.0322.

[Y5] Han S, Ye X, Astner M, Kornher T, Bhave SA, Lukin MD, Park H. Solid-state spin coherence time approaching the physical limit. *Sci Adv*. 2025;11:eadr9298. DOI: 10.1126/sciadv.adr9298.

[Y6] Barry JF, Schloss JM, Bauch E, Turner MJ, Hart CA, Pham LM, Walsworth RL. Sensitivity optimization for NV-diamond magnetometry. *Rev Mod Phys.* 2020;92(1):015004. DOI: 10.1103/RevModPhys.92.015004.

Van Schie L, Tosun M, Welter P, Givèhchi N, Lambert C-H, Gambardella P, Degen CL. Quantum magnetometry of transient signals with a time resolution of 1.1 nanoseconds. *Nat Commun.* 2025;16:822. DOI: 10.1038/s41467-025-55956-1.

[Y7] Sánchez Toural JL, Marzoa V, Bernardo-Gavito R, Pau JL, Granados D. Hands-on quantum sensing with NV⁻ centers in diamonds. *Carbon.* 2023;9(1):16. DOI: 10.3390/c9010016.

Rovny J, Gopalakrishnan S, Bleszynski Jayich AC, Maletinsky P, Demler E, de Leon NP. Nanoscale diamond quantum sensors for many-body physics. *Nat Rev Phys.* 2024;6:753–768. DOI: 10.1038/s42254-024-00775-4.